

Iowa Productivity Index Comparison

J. Stephenson

2025-10-06

Contents

```
knitr::opts_chunk$set(echo = TRUE,
  cache = TRUE,
  message = FALSE)#,
  #fig.width = 12,
  #fig.height = 12)
```

```
library("tidyverse"); theme_set(theme_bw())
library("lme4")
library("lmerTest")
library("emmeans")
library("ggResidpanel")
```

```
options(width = 200)
```

```
dir.create("fig", showWarnings = FALSE)
```

```
## Generate dataframe
ia_CPI <- read_csv("../data/raw/ia_cpi.csv") %>%
  mutate(NCCPIx100 = NCCPIall * 100,
    cpi_diff = abs(OCProdIdx - NCCPIx100),
    state = "IA")
```

```
## explore summary statistics of dataset
str(ia_CPI)
```

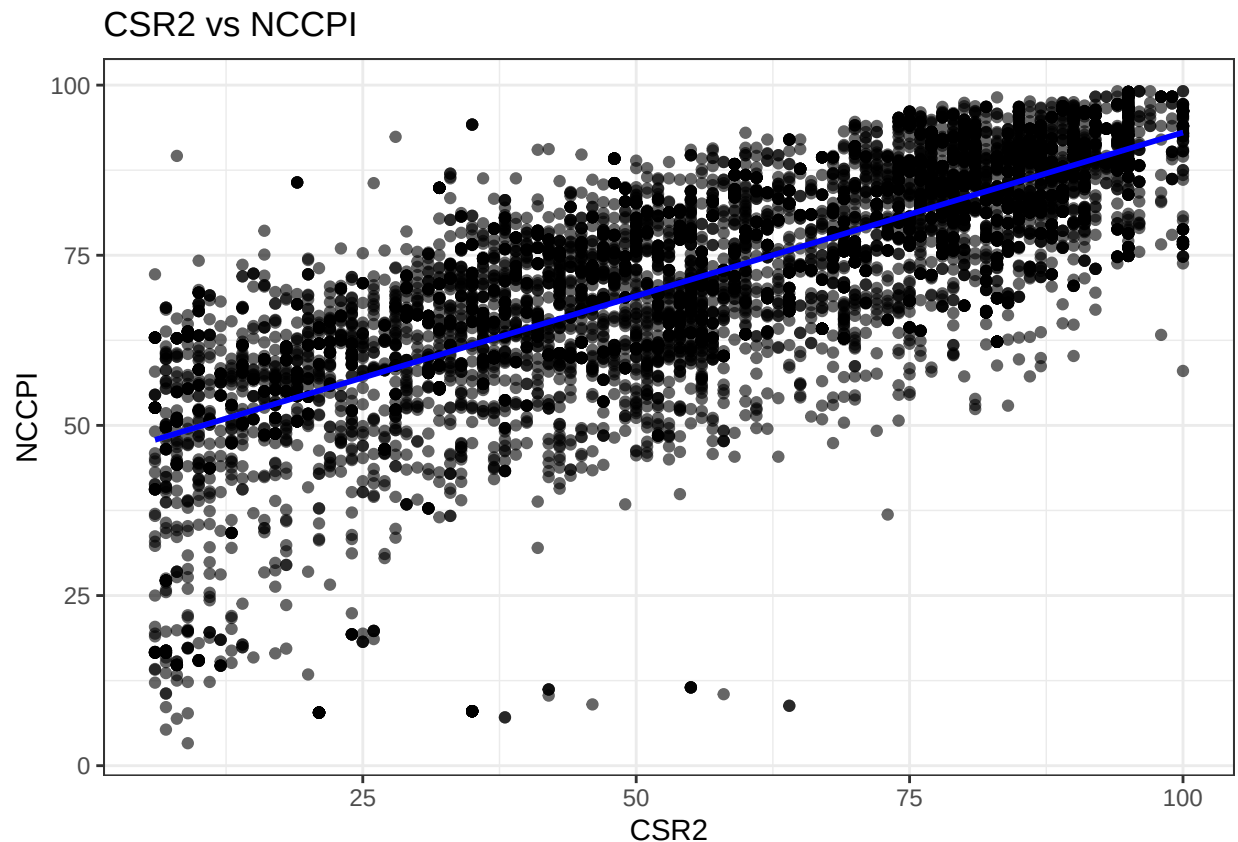
```
## tibble [9,930 x 20] (S3: tbl_df/tbl/data.frame)
## $ mukey      : num [1:9930] 402407 409105 447321 403913 401869 ...
## $ gSSURGOversion: chr [1:9930] "10/1/2024" "10/1/2024" "10/1/2024" "10/1/2024" ...
## $ MUsymbol    : chr [1:9930] "210F" "33G" "33G" "1G" ...
## $ WTDepAprJun : chr [1:9930] "NULL" "NULL" "NULL" "NULL" ...
## $ FloodFreq   : chr [1:9930] "None" "None" "None" "None" ...
## $ PondFreq    : num [1:9930] 0 0 0 0 0 0 0 0 0 0 ...
## $ DrainCls    : chr [1:9930] "Excessively drained" "Well drained" "Well drained" "Well drained" .
## $ DrainClsWet : chr [1:9930] "Excessively drained" "Well drained" "Well drained" "Well drained" .
## $ HydroGrp    : chr [1:9930] "A" "C" "C" "B" ...
## $ Hydric      : num [1:9930] 0 0 0 0 0 0 0 0 0 0 ...
## $ OCProdIdx   : num [1:9930] 6 6 6 6 6 6 6 6 6 6 ...
## $ OCprodIdxSrc : chr [1:9930] "mapunit.iacornsr (IA)" "mapunit.iacornsr (IA)" "mapunit.iacornsr (I
## $ NCCPIall    : num [1:9930] 0.122 0.141 0.142 0.166 0.166 0.166 0.166 0.166 0.166 0.167 ...
## $ RootZnDepth : chr [1:9930] "76" "150" "150" "150" ...
## $ RootZnAWS   : chr [1:9930] "64" "254" "274" "320" ...
```

```
## $ Droughty      : chr [1:9930] "1" "0" "0" "0" ...
## $ PotWetandSoil : chr [1:9930] "NULL" "NULL" "NULL" "NULL" ...
## $ NCCPIx100     : num [1:9930] 12.2 14.1 14.2 16.6 16.6 16.6 16.6 16.6 16.6 16.7 ...
## $ cpi_diff      : num [1:9930] 6.2 8.1 8.2 10.6 10.6 10.6 10.6 10.6 10.6 10.7 ...
## $ state         : chr [1:9930] "IA" "IA" "IA" "IA" ...
```

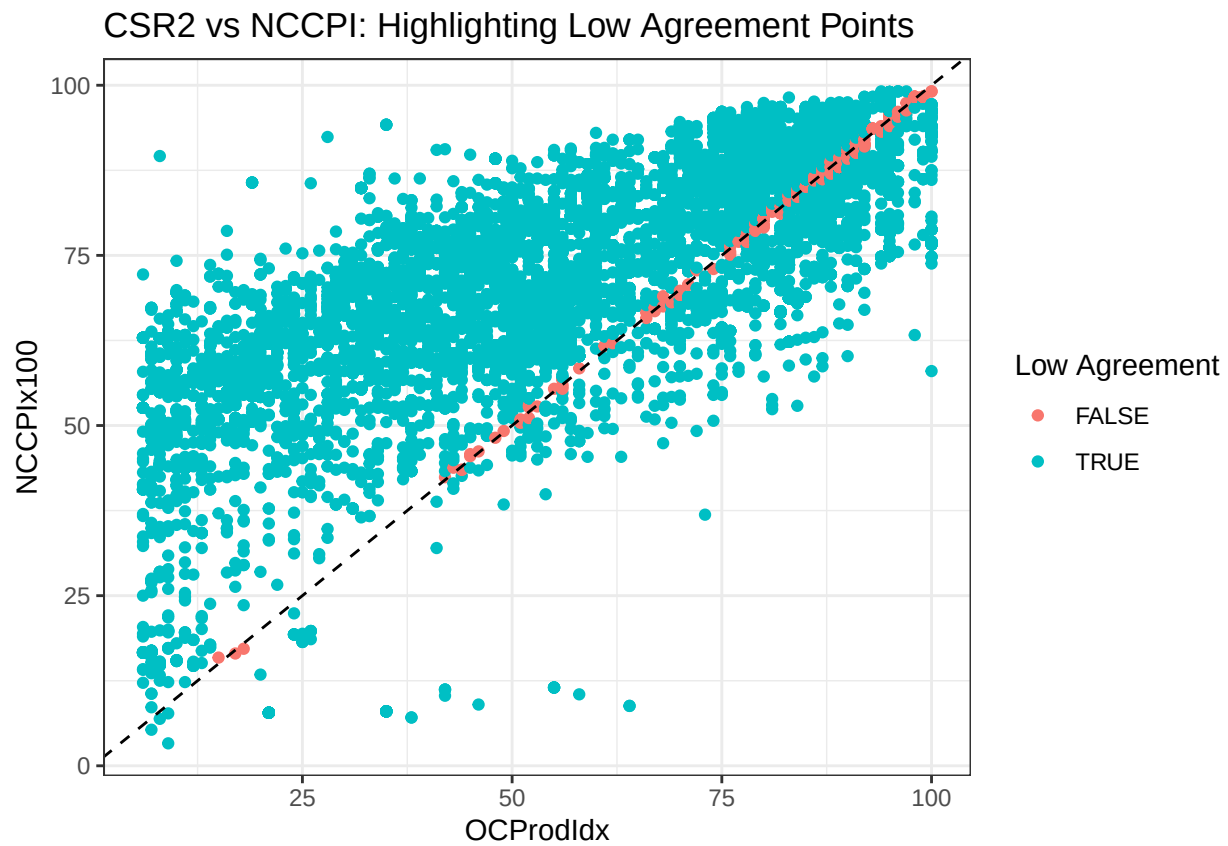
```
summary(ia_CPI)
```

```
##      mukey      gSSURGOversion      MUSymbol      WTDepAprJun      FloodFreq      Por
## Min.   : 399270 Length:9930      Length:9930      Length:9930      Length:9930      Min.
## 1st Qu.: 405895 Class :character Class :character Class :character Class :character 1st Q
## Median : 409484 Mode  :character Mode  :character Mode  :character Mode  :character Median
## Mean   : 755765                                     Mean   :0.7352                                     Mean
## 3rd Qu.: 413071                                     3rd Qu.:0.8570                                     3rd Q
## Max.   :3386724                                     Max.   :0.9910                                     Max.
##      OCProdIdx      OCprodIdxSrc      NCCPIall      RootZnDepth      RootZnAWS      Drough
## Min.   : 6.00 Length:9930      Min.   :0.0330 Length:9930      Length:9930      Length:9
## 1st Qu.: 40.00 Class :character 1st Qu.:0.6360 Class :character Class :character Class :ch
## Median : 61.00 Mode  :character Median :0.7620 Mode  :character Mode  :character Mode :ch
## Mean   : 59.36                                     Mean   :0.7352                                     Mean
## 3rd Qu.: 82.00                                     3rd Qu.:0.8570                                     3rd Q
## Max.   :100.00                                     Max.   :0.9910                                     Max.
```

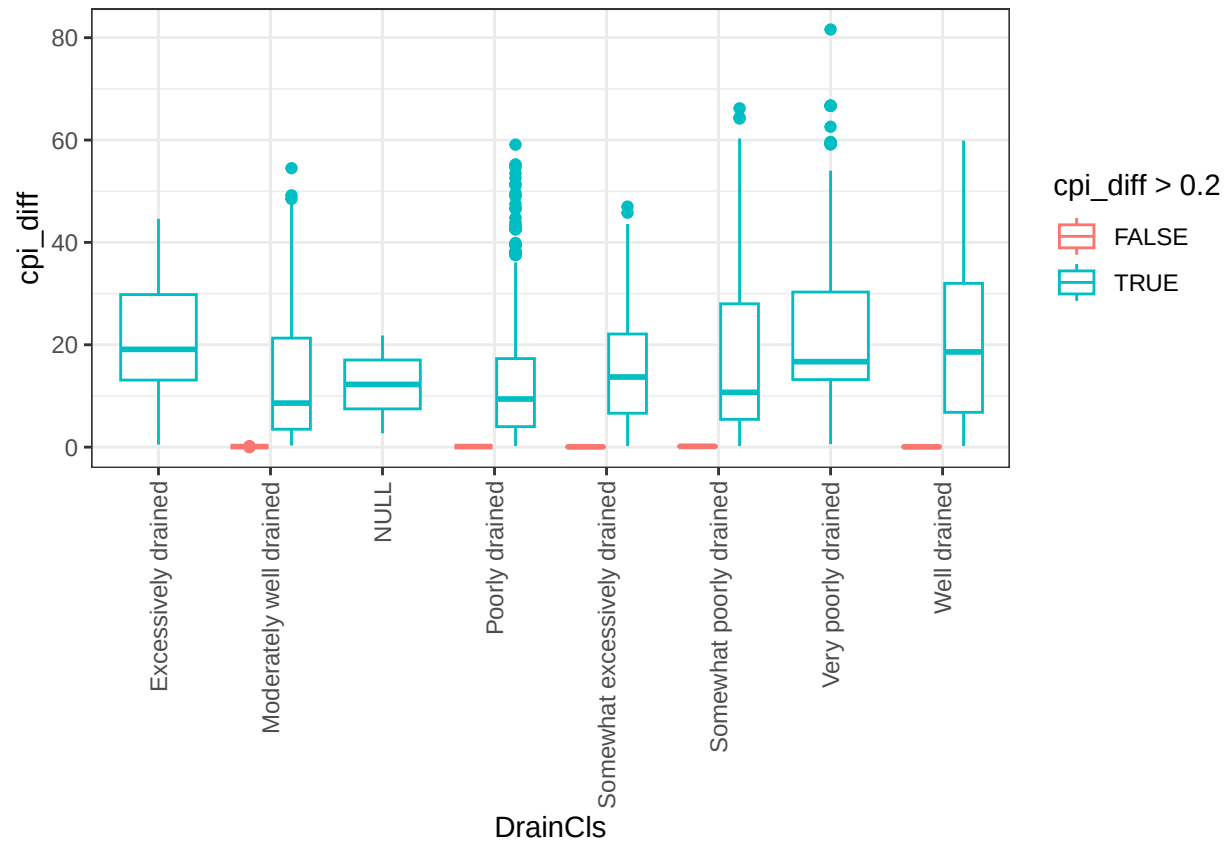
```
ggplot(ia_CPI, aes(x = OCProdIdx, y = NCCPIx100)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", color = "blue") +
  labs(title = "CSR2 vs NCCPI", x = "CSR2", y = "NCCPI")
```



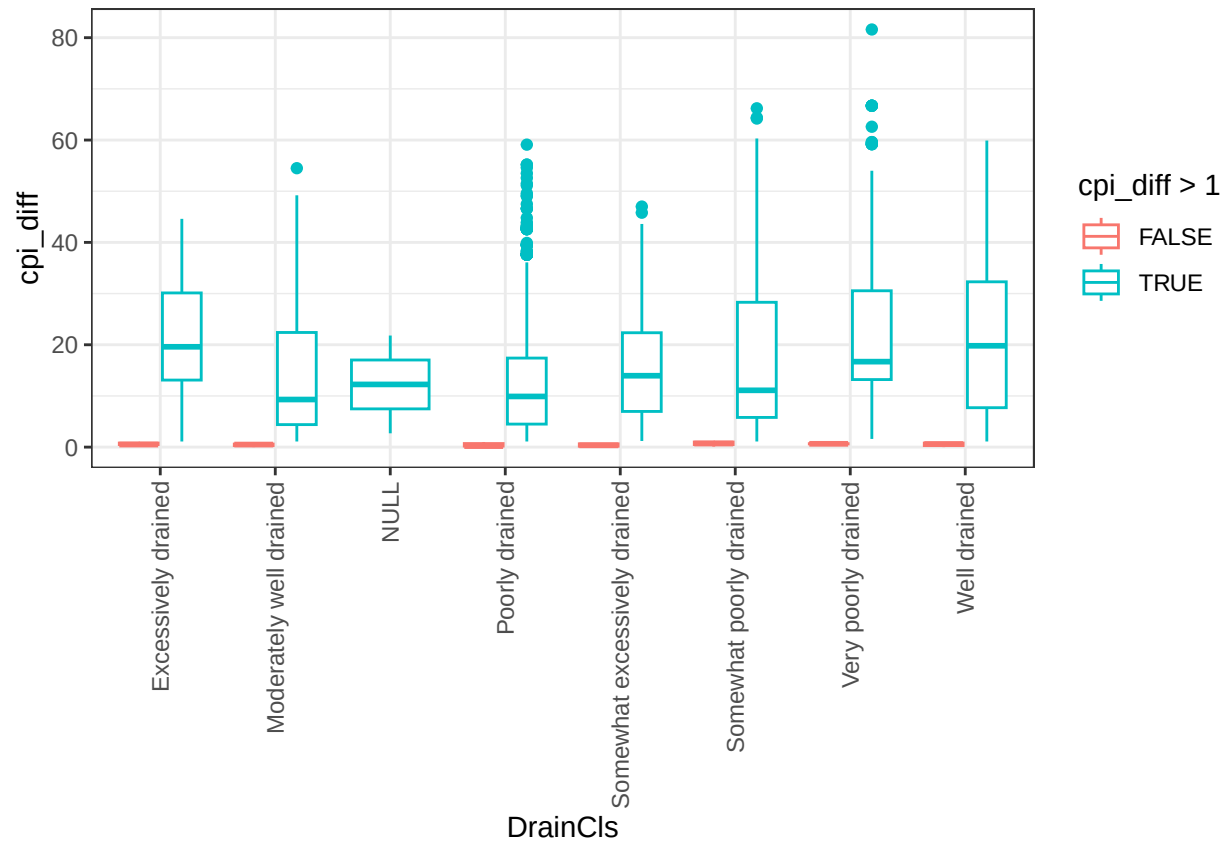
```
ggplot(ia_CPI, aes(x = OCProdIdx, y = NCCPIx100, color = cpi_diff > 1)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  labs(title = "CSR2 vs NCCPI: Highlighting Low Agreement Points",
       color = "Low Agreement")
```



```
g02 <- ggplot(ia_CPI,
  aes(x = DrainCls, y = cpi_diff, color = cpi_diff > 0.2)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5))
g02
```



```
g1 <- ggplot(ia_CPI,
  aes(x = DrainCls, y = cpi_diff, color = cpi_diff > 1)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5))
g1
```



```
g5 <- ggplot(ia_CPI,
  aes(x = DrainCls, y = cpi_diff, color = cpi_diff > 5)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5))
g5
```

